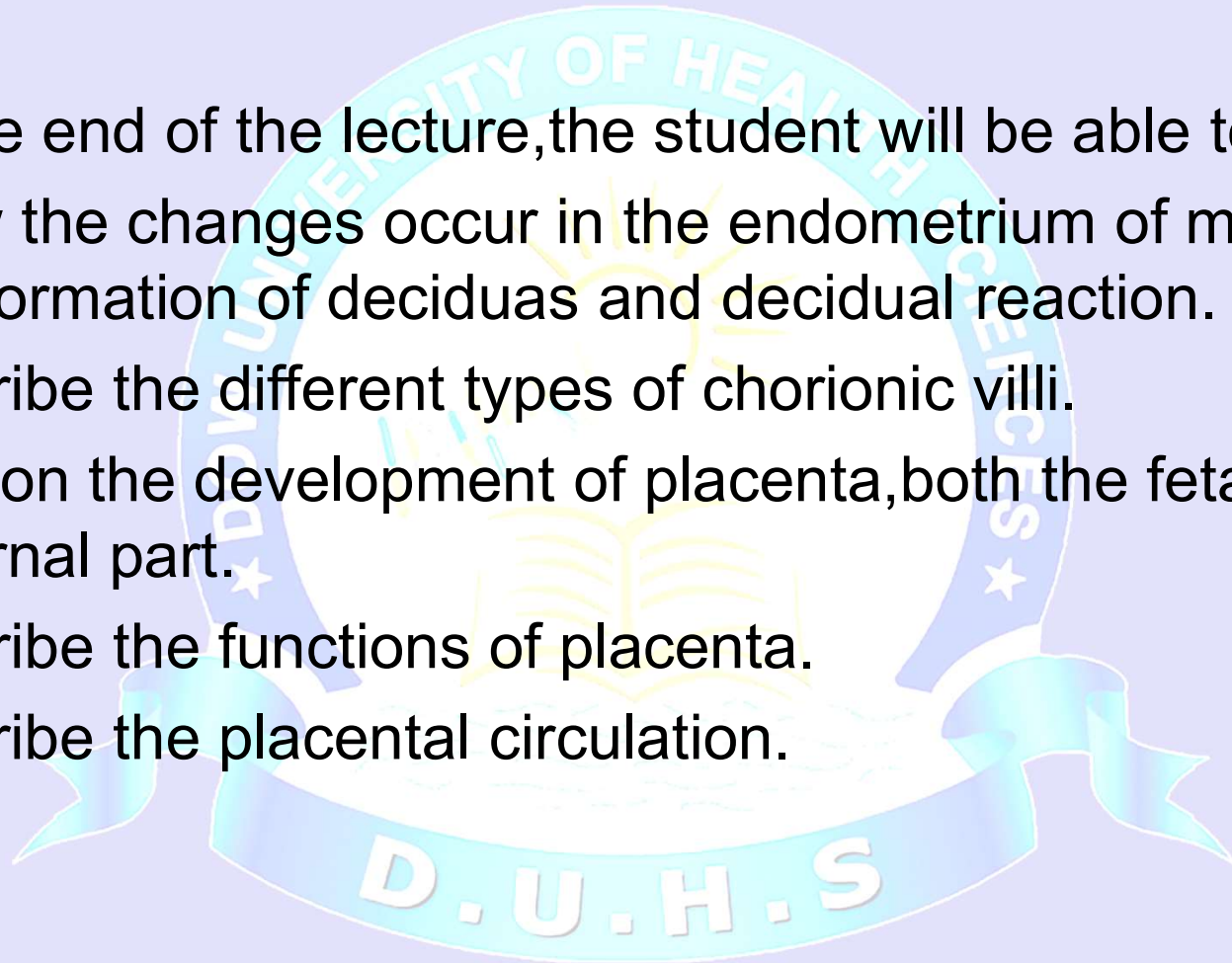




Placenta

LEARNING OBJECTIVE.

- By the end of the lecture, the student will be able to:
- Know the changes that occur in the endometrium of mother with formation of deciduas and decidual reaction.
- Describe the different types of chorionic villi.
- Mention the development of placenta, both the fetal and maternal part.
- Describe the functions of placenta.
- Describe the placental circulation.



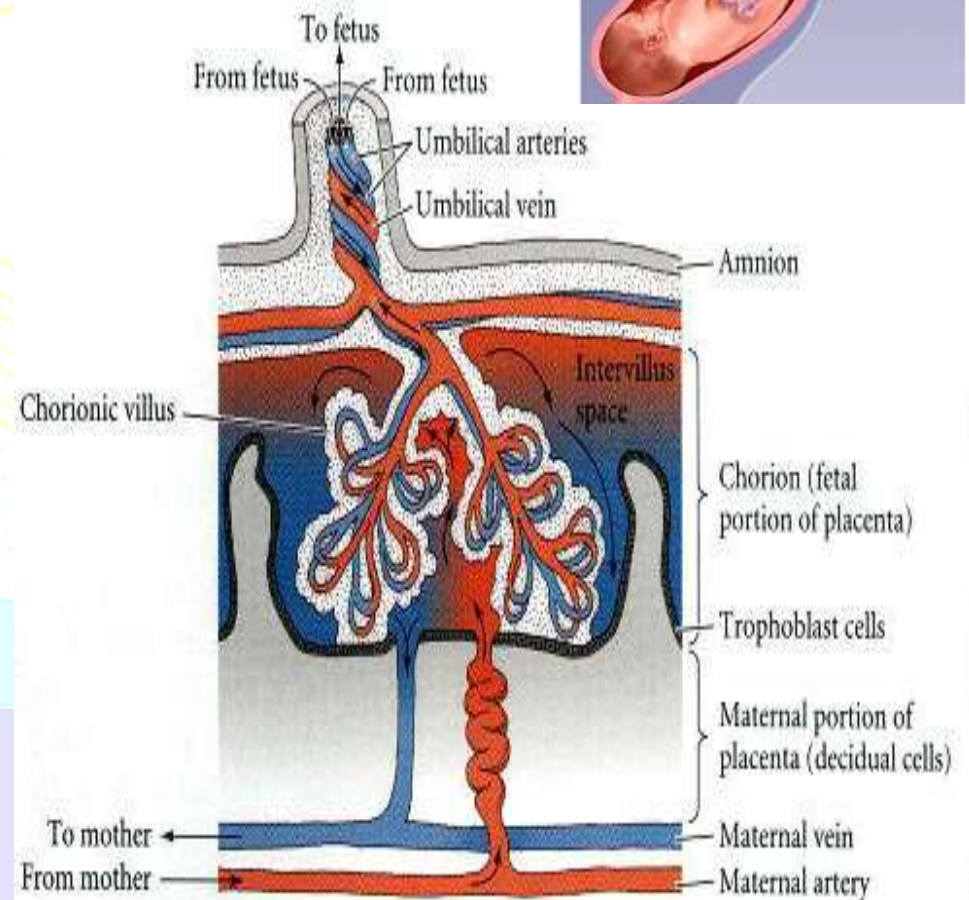
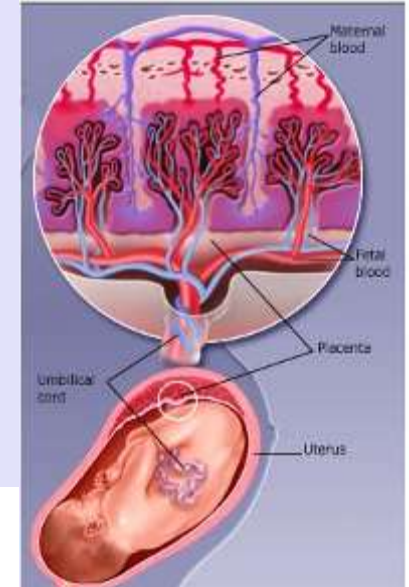
The Placenta

- Primary site of nutrient and gas exchange.

Components:

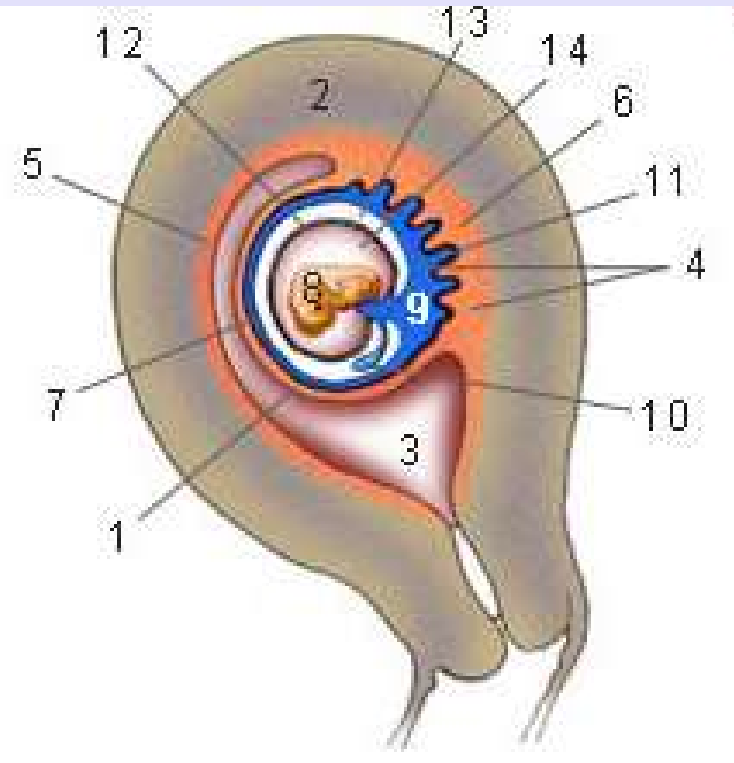
- A **fetomaternal organ** having two components:
 - Fetal component : **the villous chorion**.
 - Maternal Component: the decidua basalis.

Normal Placenta



The Decidua

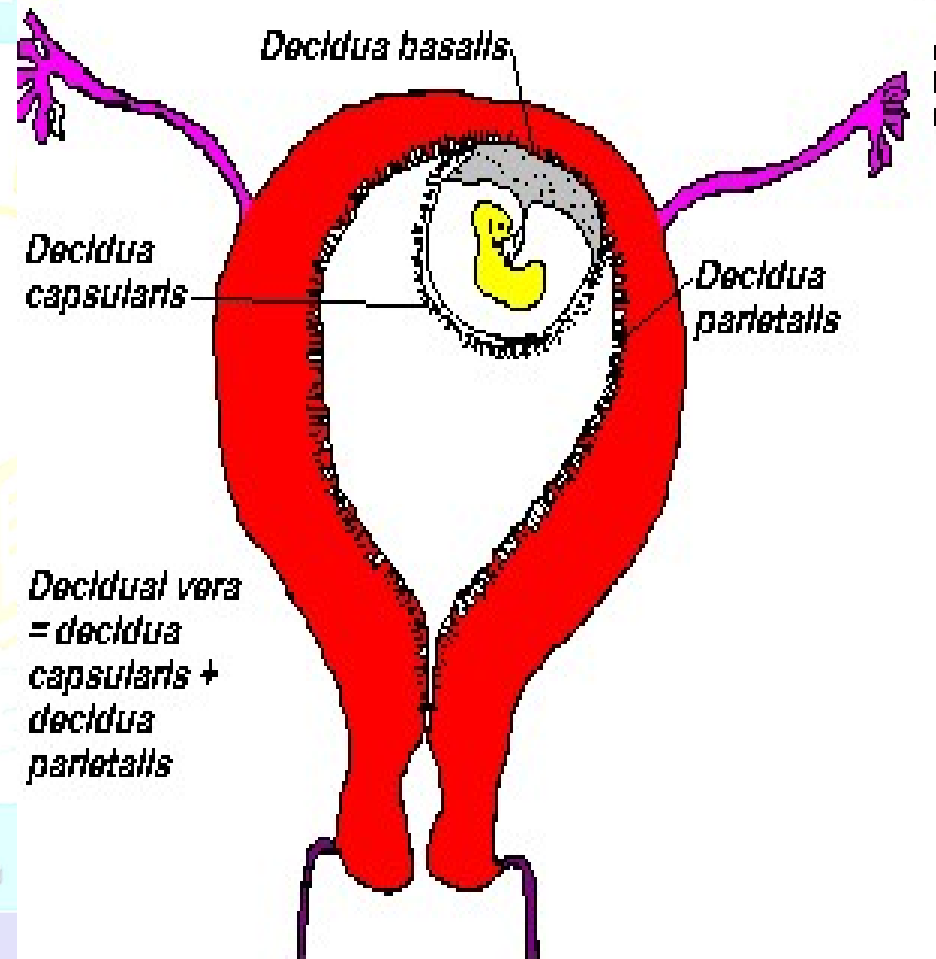
- Decidua refers to gravid endometrium, the functional layer of the endometrium in a pregnant women.
- Separates from the uterus after childbirth.



1. Decidua capsularis
2. Uterine wall
3. Uterine cavity
4. Placenta
5. Decidua parietalis
6. Decidua basalis

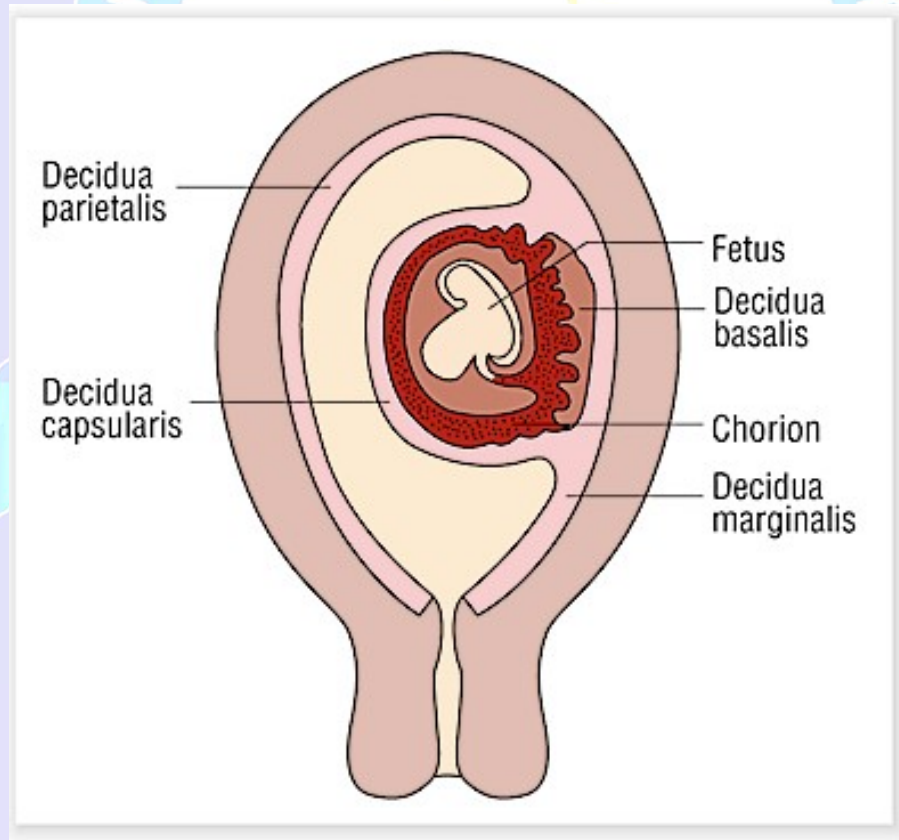
Three regions of decidua

- **Decidua Basalis:** Part of decidua deep to the conceptus.
- **Decidua Capsularis:** Superficial part of the decidua overlying the conceptus.
- **Decidua Parietalis:** all remaining parts.

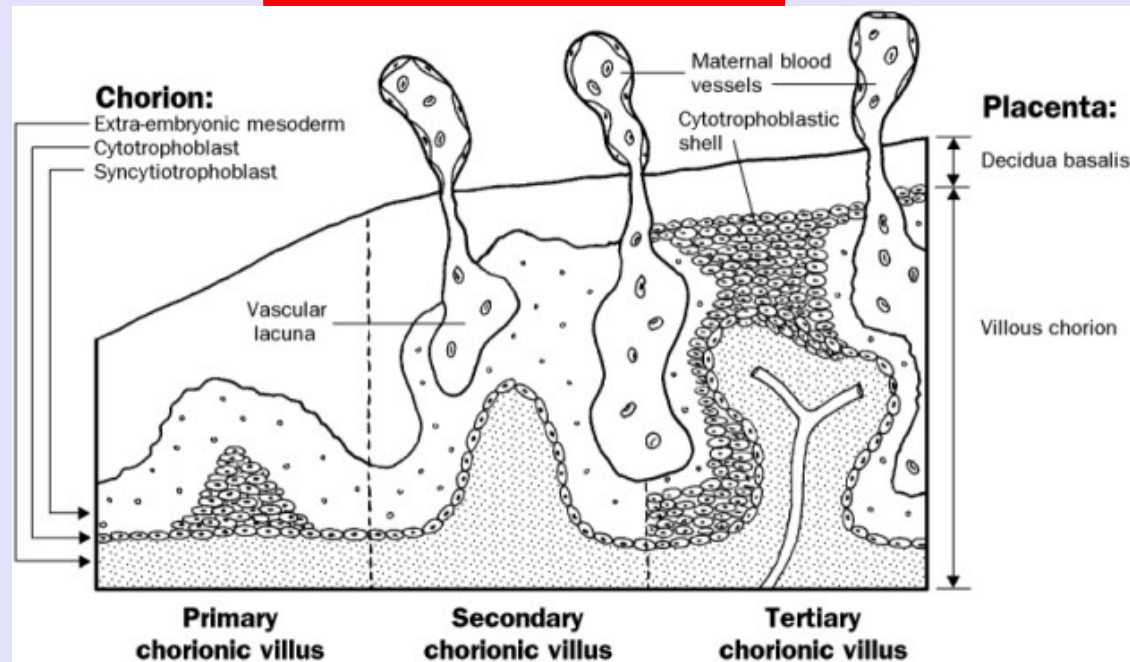


Shape

- Discoid, 15 to 20 cm in the diameter, 2 to 3 cm thickness, 500 to 600 gm weight.



Structure



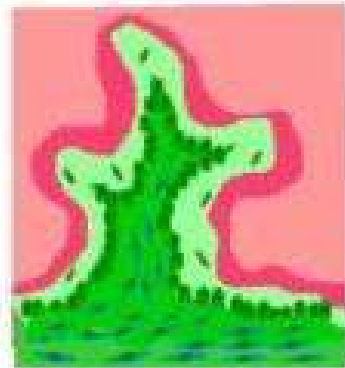
- **Primary chorionic villi** are solid outgrowths of cytotrophoblast that protrude into the syncytiotrophoblast.
- **Secondary chorionic villi** have a core of loose connective tissue, which grows into the primary villi about the third week of development.

Structure

- Tertiary chorionic villi contain **embryonic blood vessels** that develop from mesenchymal cells in the loose connective tissue core.
- These blood vessels connect up with vessels that develop in the chorion and connecting stalk and begin to circulate embryonic blood about the **third week** of development.



Primary villi consist of syncytiotrophoblast and cytotrophoblast day 13



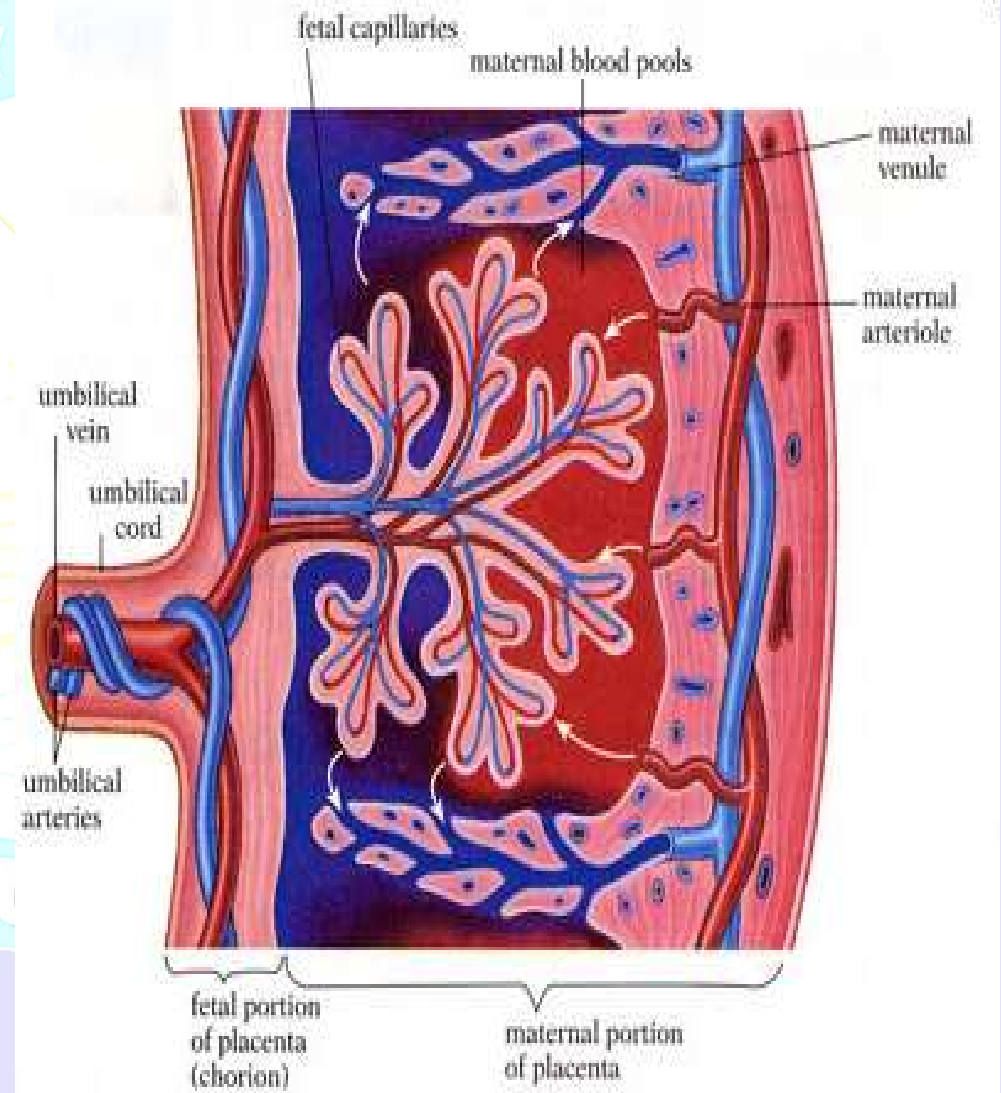
Secondary villi have a core of extra-embryonic mesoderm day 16



Tertiary villi have blood vessels in the mesoderm day 21

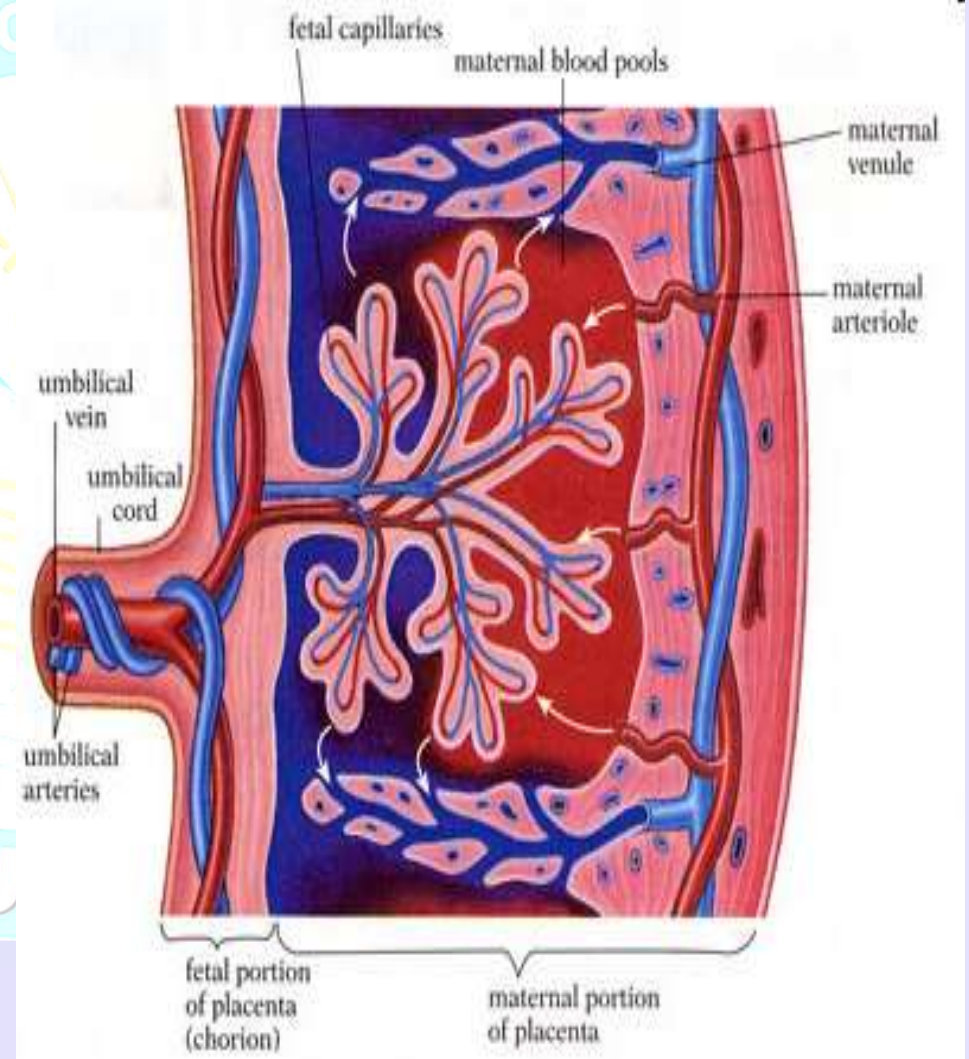
The fetal part of placenta

- fetal portion of the placenta consists of the villi of the **chorion frondosum**;
- These villi are suspended in the intervillous space, and bathed in maternal blood.
- It is the **chorionic portion of the placenta**, containing the fetal blood vessels, from which the umbilical cord arises.



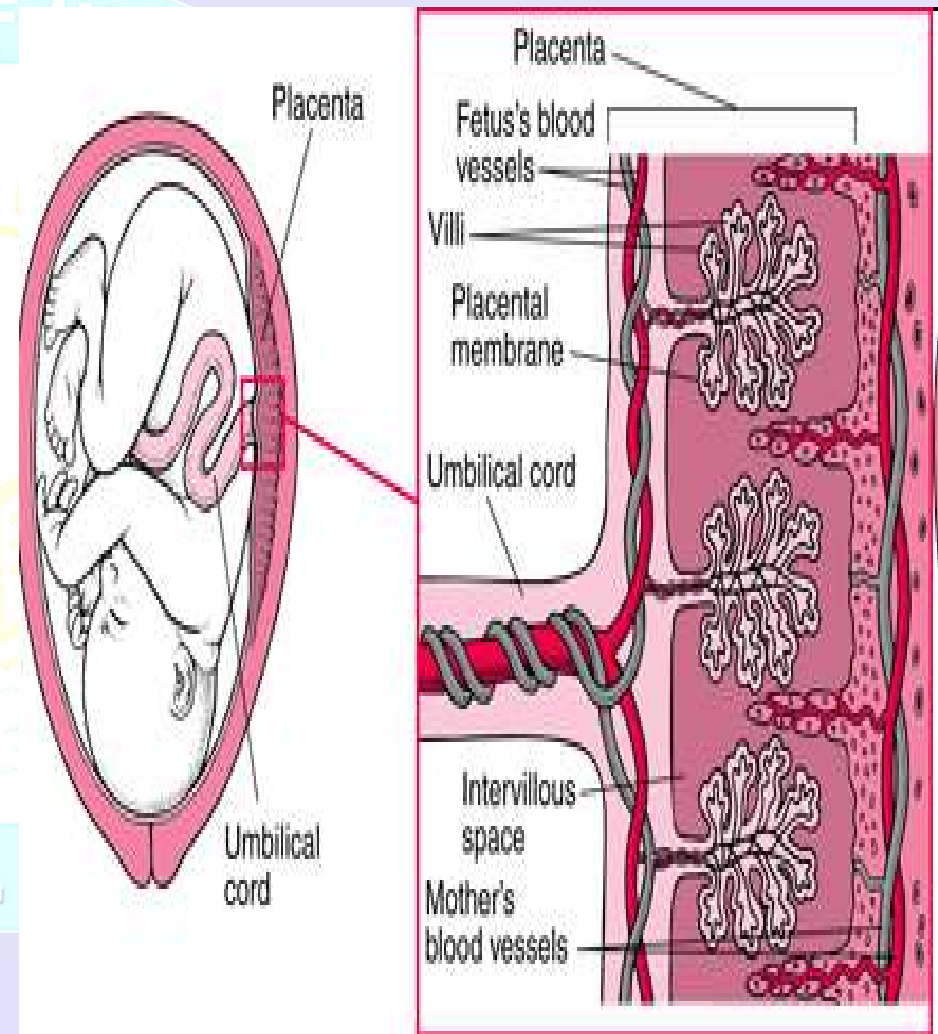
Maternal Portion.

- Formed by the decidua basalis containing the **intervillous space.**
- From the maternal side the placenta has a wrinkled surface, characterized by the **cotyledons.**



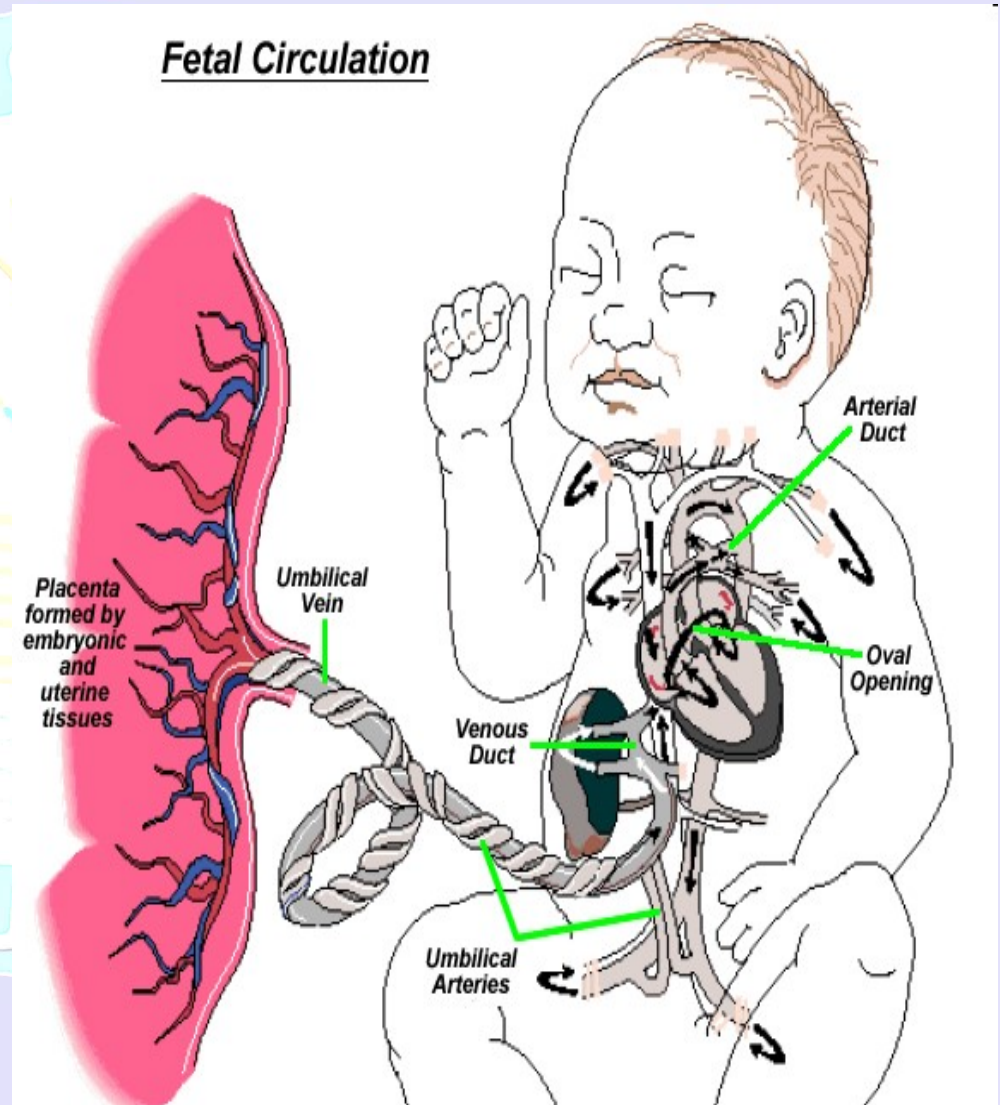
Intervillous Space

- **Intervillous space** of the placenta contain maternal blood.
- Derived from the **lacunae** that developed in the syncytiotrophoblast during the second week.
- This large blood – filled space results from the coalescence and enlargement of the lacunar networks.



Placental Circulation

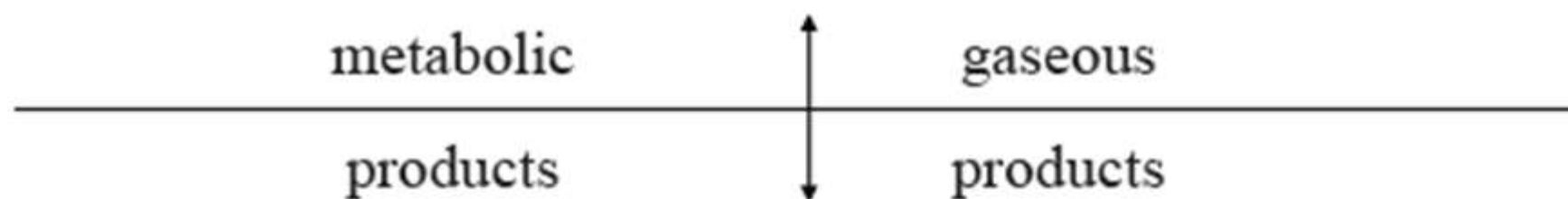
- Circulation of blood through the placenta during intrauterine life, serving fetal needs for aeration, absorption, and excretion.
- Placental blood flow is increased at term and amounts to **500 ml/min (80% of the uterine perfusion)**.



Placental circulation

fetal placental circulation
(within the villus)

umbilical A. → → arterio-capillary-venous system → → umbilical V.



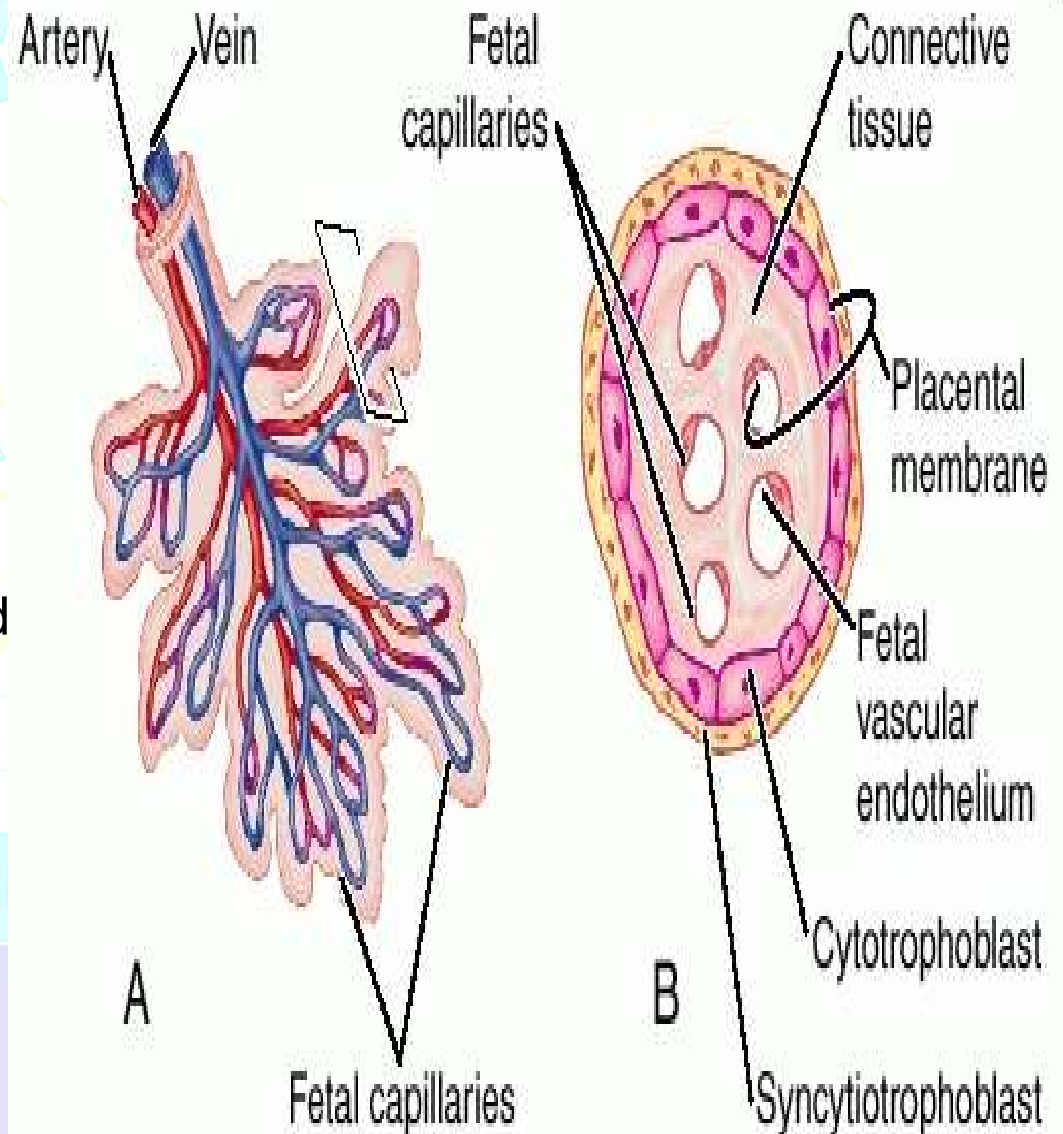
spiral A. → intervillous space → endometrial V.

maternal placental circulation

intrauterine growth retardation (IUGR)

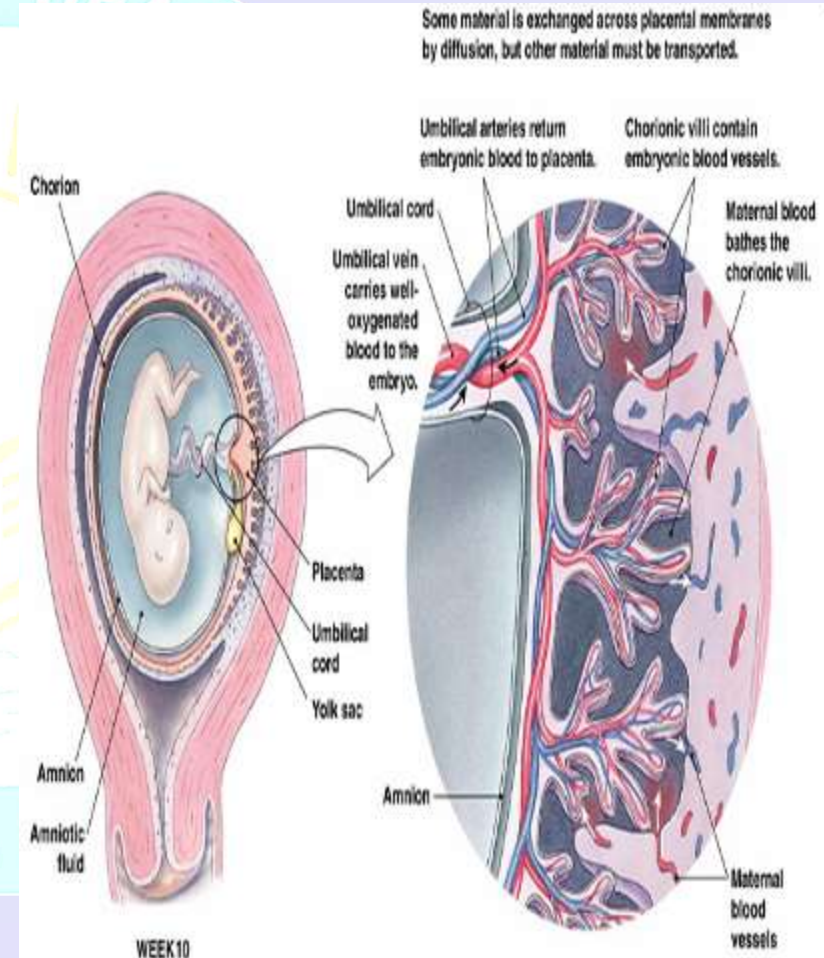
The placental barrier

- The placental barrier is composed of structures that separate the maternal and the fetal blood.
- **In the first trimester** it consists of:
 - the syncytiotrophoblast,
 - the cytotrophoblast (Langhans' cells),
 - the villus mesenchyma and
 - the fetal capillary walls.
- **During the 4th month** the cytotrophoblast disappears from the villus wall



Functions of the placenta

- Metabolism.(synthesis of glycogen).
- Transport of gases and nutrients.
- Endocrine secretion (human chorionic gonadotrophin - hCG).
- Transport of maternal antibodies.
- Exchange of electrolytes.
- Excretion of waste products ,example urea and uric acid.



References

- The Developing Human
Clinically Oriented Embryology.
Eighth Edition. Keith .L. Moore.
T.V.N.Persaud.

